Understanding the DFT: From Time Domain to Frequency Domain

0. Foundations: Frequency, Period, and Magnitude

Before diving into the Discrete Fourier Transform (DFT), we need a clear understanding of three basic signal properties:

- **frequency**,
- **period**,
- and **magnitude** (or amplitude).

These concepts are the building blocks of the frequency domain.

Period and Frequency

A **periodic signal** repeats itself after a fixed time interval called the **period** T (measured in seconds).

The **frequency** f is the number of cycles that occur per second, measured in **Hertz (Hz)**. They are inversely related:

$$f = \frac{1}{T}, \qquad T = \frac{1}{f}.$$

For example, a sine wave with period $T = 0.02\,\text{s}$ has frequency $f = 50\,\text{Hz}$ – it completes 50 cycles every second.

Sometimes we use **angular frequency** ω, measured in radians per second:

$$\omega = 2\pi f = \frac{2\pi}{T}.$$

This notation simplifies many formulas, e.g., a pure sinusoid can be written as $\sin(\omega t)$ or $\cos(\omega t)$.

Magnitude (Amplitude)

The **magnitude** (or **amplitude**) of a signal describes its strength.

For a sine wave $A\sin(2\pi f t)$,

- the amplitude A is the peak deviation from zero.

In the frequency domain, the DFT shows **how much** of each frequency is present – that "how much" is the magnitude.

- A larger amplitude → a taller peak in the spectrum.
- A real sine wave of amplitude A produces two DFT peaks, each of magnitude $A/2$ (because a real sine is the sum of two complex exponentials, as explained later).

Why These Matter for the DFT

The DFT converts a time-domain signal into a set of complex coefficients y_k. For each frequency index k:

- The **magnitude** $|y_k|$ tells you the amplitude of that frequency component.
- The **phase** (angle of y_k) tells you the time shift of that component.
- The **frequency** associated with index k is $f_k = k \cdot \frac{f_s}{N}$, where f_s is the sampling rate and N the number of samples.

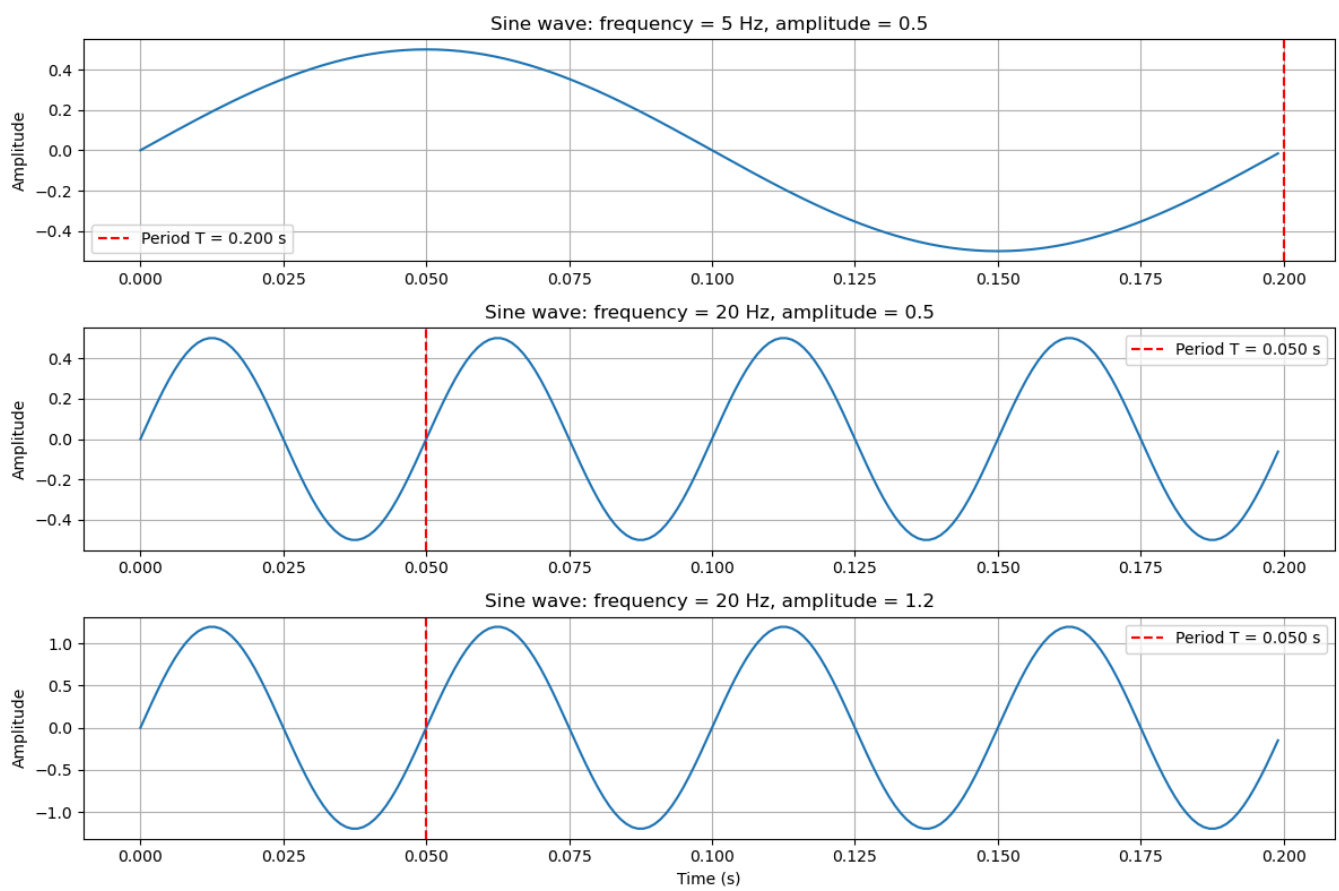
Understanding period, frequency, and magnitude is essential to correctly interpret DFT plots and to avoid common pitfalls like aliasing (where a high frequency masquerades as a low one due to insufficient sampling).

> **In the Quantum Fourier Transform (QFT):** The same fundamental ideas of frequency and magnitude appear, but in a quantum mechanical setting. QFT acts on probability amplitudes encoded in qubit states, revealing periodic structures in quantum algorithms (e.g., Shor's algorithm).

0.1 Seeing Period, Frequency, and Magnitude in Action

```python
import numpy as np
import matplotlib.pyplot as plt

# Common parameters
fs = 1000          # sampling frequency (Hz)
duration = 0.2     # short duration to see individual cycles (seconds)
t = np.linspace(0, duration, int(fs * duration), endpoint=False)

# 1) Low frequency, low magnitude
f1, A1 = 5, 0.5
sine1 = A1 * np.sin(2 * np.pi * f1 * t)

# 2) Higher frequency, same magnitude
f2, A2 = 20, 0.5
sine2 = A2 * np.sin(2 * np.pi * f2 * t)

# 3) Same frequency as (2) but larger magnitude
f3, A3 = 20, 1.2
sine3 = A3 * np.sin(2 * np.pi * f3 * t)

# Plot all three
plt.figure(figsize=(12, 8))

plt.subplot(3, 1, 1)
plt.plot(t, sine1)
plt.title(f'Sine wave: frequency = {f1} Hz, amplitude = {A1}')
plt.ylabel('Amplitude')
plt.grid(True)
# Mark one period
T1 = 1 / f1
plt.axvline(x=T1, color='r', linestyle='--', label=f'Period T = {T1:.3f} s')
plt.legend()

plt.subplot(3, 1, 2)
plt.plot(t, sine2)
plt.title(f'Sine wave: frequency = {f2} Hz, amplitude = {A2}')
plt.ylabel('Amplitude')
plt.grid(True)
T2 = 1 / f2
```

```
plt.axvline(x=T2, color='r', linestyle='--', label=f'Period T = {T2:.3f} s')
plt.legend()

plt.subplot(3, 1, 3)
plt.plot(t, sine3)
plt.title(f'Sine wave: frequency = {f3} Hz, amplitude = {A3}')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.grid(True)
plt.axvline(x=T2, color='r', linestyle='--', label=f'Period T = {T2:.3f} s')
plt.legend()

plt.tight_layout()
plt.show()
```



1. Introduction

Signals can be viewed in two complementary ways:

- **Time domain** – how the signal changes over time.
- **Frequency domain** – which frequencies make up the signal and with what amplitudes.

The **Discrete Fourier Transform (DFT)** converts a sampled time-domain signal into its frequency components. In this notebook we will build intuition by experimenting with simple signals.

Quick Math Review

The DFT maps a sequence of N complex numbers $x_0, x_1, \ldots, x_{N-1}$ to another sequence $y_0, y_1, \ldots, y_{N-1}$:

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \, e^{-2\pi i \, jk/N}$$

- **Inverse DFT** uses $+$ sign in exponent and same normalization.
- **Complexity:** Naïve computation takes $O(N^2)$ operations.
- **Fast Fourier Transform (FFT):** $O(N \log N)$ classical algorithm.

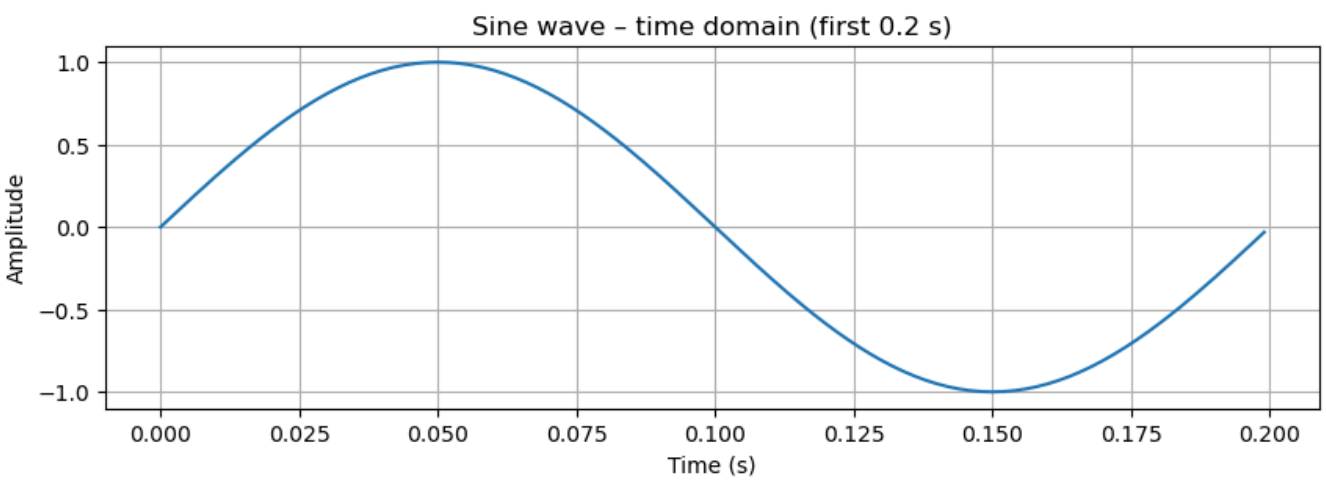
---

2. Our First Signal: A Pure Sine Wave

We generate a sine wave of frequency $f_0 = 5\,\text{Hz}$, sampled at $f_s = 1000\,\text{Hz}$ for 1 second.

```
import numpy as np
import matplotlib.pyplot as plt
import ipywidgets as widgets
from IPython.display import display, clear_output
%matplotlib inline
```

```
fs = 1000           # sampling frequency (Hz)
T = 1.0             # duration (seconds)
N = int(fs * T)     # number of samples
t = np.linspace(0, T, N, endpoint=False)   # time vector

f0 = 5              # frequency of the sine (Hz)
sine = np.sin(2 * np.pi * f0 * t)

# Plot the first 0.2 seconds
plt.figure(figsize=(10, 3))
plt.plot(t[:200], sine[:200])
plt.title('Sine wave — time domain (first 0.2 s)')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.grid(True)
plt.show()
```

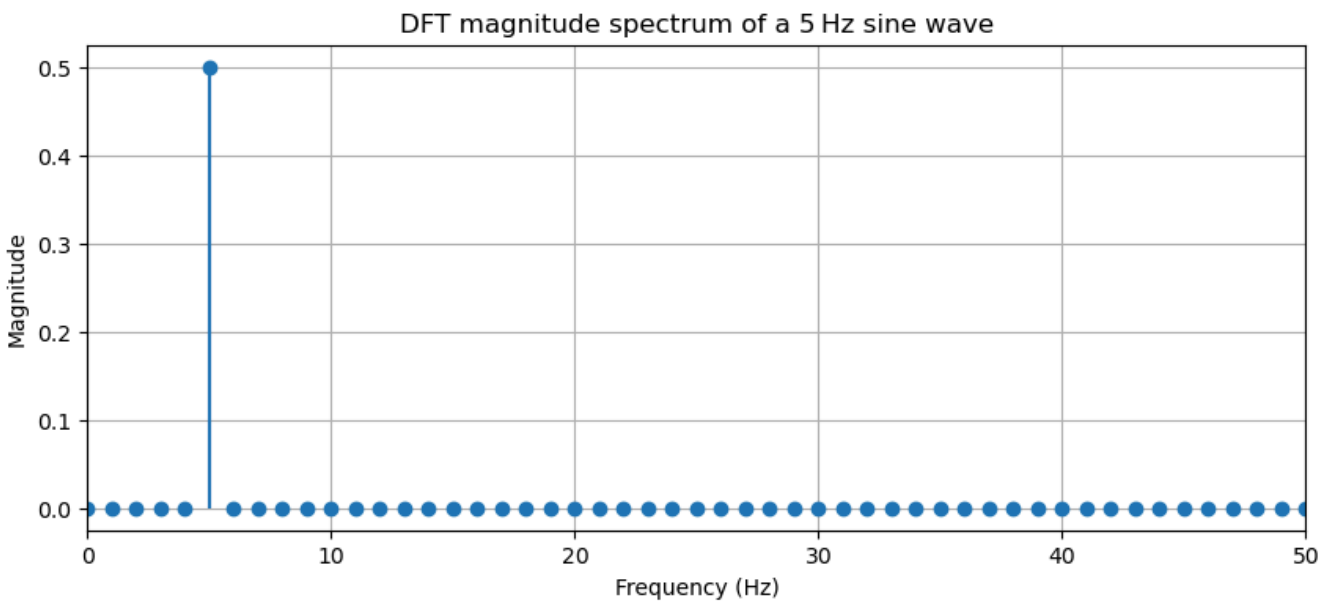


2.1 Compute and plot the DFT

We use `numpy.fft.fft` and normalise the magnitude by the number of samples `N` to obtain the actual amplitude of the sinusoidal components.

> **Important:** For a real sine wave $\sin(2\pi f_0 t)$, the DFT yields two peaks: one at $+f_0$ and one at $-f_0$ (or, in the one-sided spectrum, reflected). Each peak has magnitude $A/2$ where A is the sine amplitude. We will see why.

```
X = np.fft.fft(sine)
# One-sided spectrum (positive frequencies only)
freq = np.fft.fftfreq(N, 1/fs)[:N//2]
mag = np.abs(X[:N//2]) / N   # normalised magnitude

plt.figure(figsize=(10, 4))
plt.stem(freq, mag, basefmt=" ")
plt.xlim(0, 50)   # show up to 50 Hz
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude')
plt.title('DFT magnitude spectrum of a 5 Hz sine wave')
plt.grid(True)
plt.show()
```

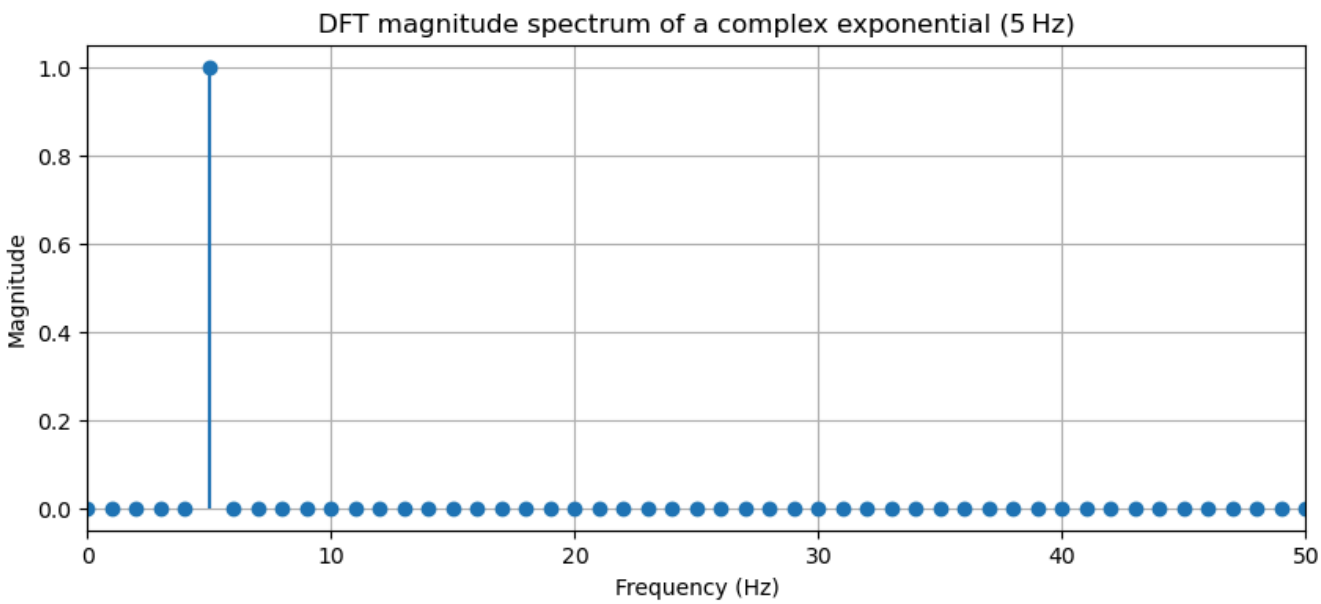


Observation: The peak is at 5 Hz with magnitude **0.5**.
Why 0.5? Because $\sin(2\pi f_0 t) = \frac{e^{j2\pi f_0 t} - e^{-j2\pi f_0 t}}{2j}$.
Each complex exponential has amplitude $1/2$. The DFT (normalised) shows that half-amplitude contribution.

3. Complex Exponential vs. Real Sine

A complex exponential $e^{j2\pi f_0 t}$ is a single rotating phasor. Its DFT should show **one peak** of magnitude **1**.

```
complex_exp = np.exp(1j * 2 * np.pi * f0 * t)

Xc = np.fft.fft(complex_exp)
mag_c = np.abs(Xc[:N//2]) / N

plt.figure(figsize=(10, 4))
plt.stem(freq, mag_c, basefmt=" ")
plt.xlim(0, 50)
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude')
plt.title('DFT magnitude spectrum of a complex exponential (5 Hz)')
plt.grid(True)
plt.show()
```



Now we see a single peak of magnitude **1** at 5 Hz.
This illustrates that a real sinusoid is the sum of two complex exponentials (positive and negative frequency), hence the factor $1/2$.

4. The Clock Mechanism: Visualising Period and Frequency

Imagine a hand on a clock rotating with constant angular speed. The projection of that hand onto the horizontal axis gives a cosine wave, onto the vertical axis a sine wave. The **angular**

frequency $\omega = 2\pi f$ determines how many full rotations (periods) occur per second.

Let's create a small animation to see the relationship between the rotating phasor and the resulting sinusoidal signal.

```python
from matplotlib.animation import FuncAnimation
from IPython.display import HTML

# Parameters for the animation
f_anim = 1          # 1 Hz (one full rotation per second)
duration = 2.0      # seconds
frames = 50         # number of frames
time_anim = np.linspace(0, duration, frames)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 5))
ax1.set_xlim(-1.2, 1.2)
ax1.set_ylim(-1.2, 1.2)
ax1.set_aspect('equal')
ax1.grid(True)
ax1.set_title('Rotating phasor')
ax1.set_xlabel('Real')
ax1.set_ylabel('Imag')
line, = ax1.plot([], [], 'o-', lw=2)
trace_x, = ax1.plot([], [], 'r--', lw=1)

ax2.set_xlim(0, duration)
ax2.set_ylim(-1.2, 1.2)
ax2.grid(True)
ax2.set_title('Cosine wave (projection on real axis)')
ax2.set_xlabel('Time (s)')
ax2.set_ylabel('Amplitude')
cos_line, = ax2.plot([], [], 'b-', lw=2)

time_data = []
cos_data = []

def init():
    line.set_data([], [])
    trace_x.set_data([], [])
    cos_line.set_data([], [])
    return line, trace_x, cos_line

def update(frame):
    t = time_anim[frame]
```

```
    angle = 2 * np.pi * f_anim * t
    x = np.cos(angle)
    y = np.sin(angle)
    line.set_data([0, x], [0, y])

    # Draw the x-projection (vertical line from phasor tip to real axis)
    trace_x.set_data([x, x], [0, y])

    time_data.append(t)
    cos_data.append(x)
    if len(time_data) > 100:
        time_data.pop(0)
        cos_data.pop(0)
    cos_line.set_data(time_data, cos_data)
    return line, trace_x, cos_line

anim = FuncAnimation(fig, update, frames=frames, init_func=init, blit=True, repeat=True)
HTML(anim.to_jshtml())
```

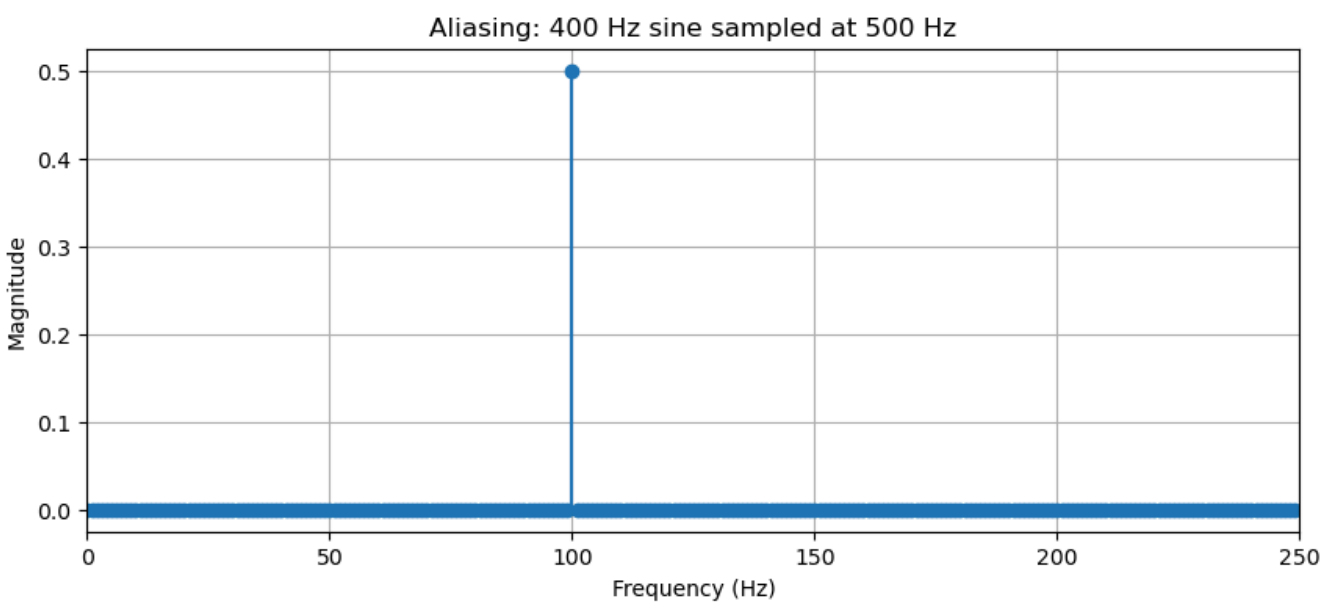
Run the cell above to see the animation. The rotating phasor (blue arrow) sweeps a full circle every $1/f$ seconds. Its projection onto the real axis gives a cosine wave.
Exercise: Change `f_anim` to 2 Hz and re-run. What changes? The period becomes 0.5 seconds – the wave compresses.

5. Effect of Sampling Rate and the Nyquist Theorem

The sampling frequency f_s must be at least twice the highest frequency present in the signal (Nyquist rate). If we sample too slowly, high frequencies **alias** to lower frequencies.

Let's demonstrate aliasing by generating a 400 Hz sine wave but sampling it at $f_s = 500$ Hz (Nyquist = 250 Hz, too low).

```
fs_low = 500
t_low = np.linspace(0, T, int(fs_low * T), endpoint=False)
f_alias = 400
sine_alias = np.sin(2 * np.pi * f_alias * t_low)

X_alias = np.fft.fft(sine_alias)
freq_low = np.fft.fftfreq(len(t_low), 1/fs_low)[:len(t_low)//2]
mag_alias = np.abs(X_alias[:len(t_low)//2]) / len(t_low)

plt.figure(figsize=(10, 4))
plt.stem(freq_low, mag_alias, basefmt=" ")
plt.xlim(0, 250)
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude')
plt.title(f'Aliasing: 400 Hz sine sampled at {fs_low} Hz')
plt.grid(True)
plt.show()
```



The peak appears not at 400 Hz but at $f_s - 400 = 100$ Hz – the alias of the 400 Hz component.
Rule of thumb: Always choose $f_s > 2 \times f_{\max}$.

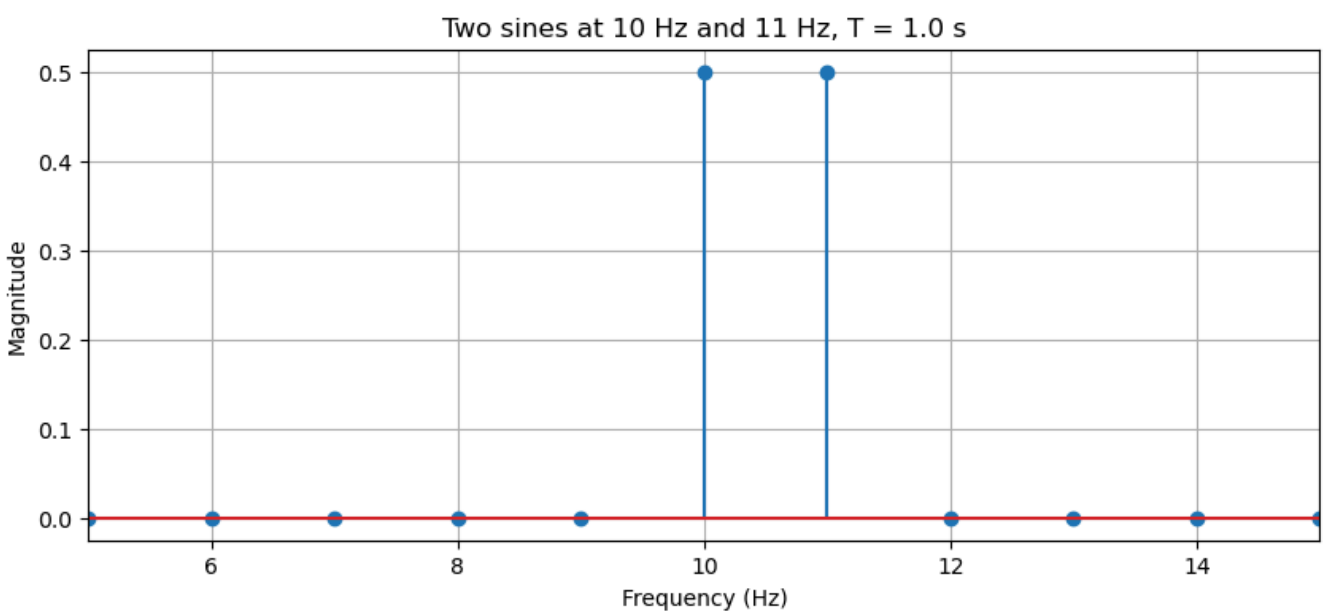
6. Frequency Resolution: Duration Matters

The DFT splits the frequency range $[0, f_s)$ into N bins. The spacing between bins (frequency

resolution) is $\Delta f = f_s/N = 1/T$.
If two frequencies are closer than Δf, they may appear as a single broad peak.

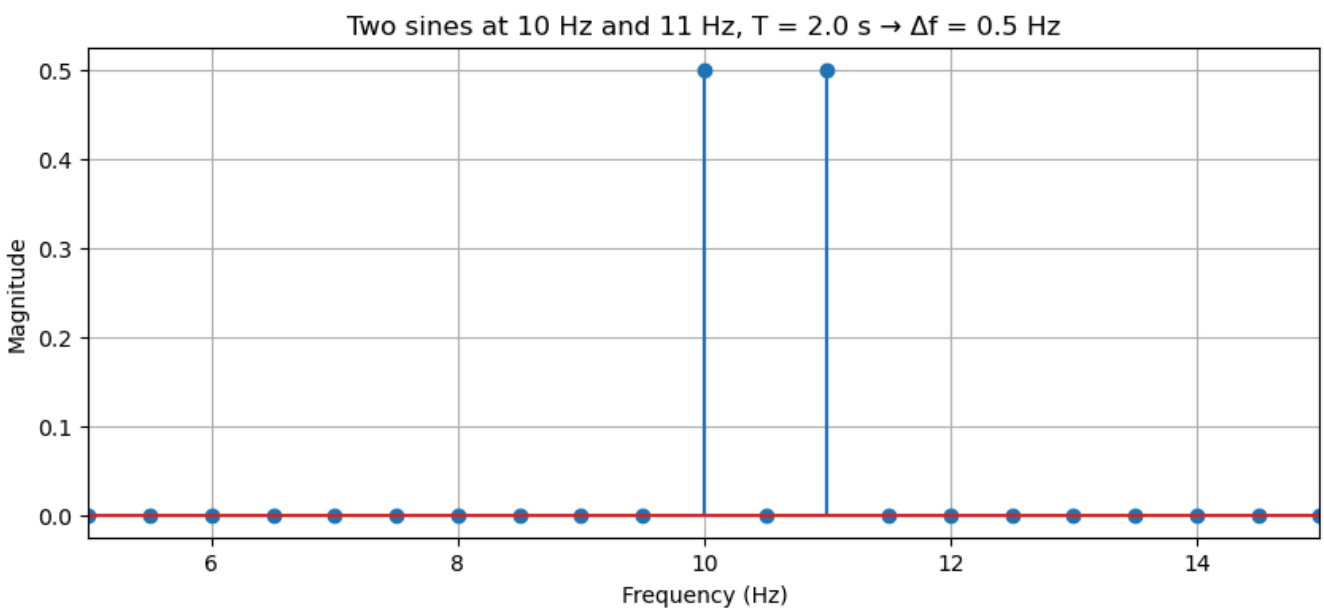
Example: Two sine waves at 10 Hz and 11 Hz, duration $T = 1\,\text{s} \to \Delta f = 1\,\text{Hz}$, so they are just resolvable. Let's see.

```
f1, f2 = 10, 11
two_sines = np.sin(2*np.pi*f1*t) + np.sin(2*np.pi*f2*t)

X_two = np.fft.fft(two_sines)
mag_two = np.abs(X_two[:N//2]) / N

plt.figure(figsize=(10, 4))
plt.stem(freq, mag_two)   # use plot instead of stem for clarity
plt.xlim(5, 15)
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude')
plt.title(f'Two sines at {f1} Hz and {f2} Hz, T = {T} s')
plt.grid(True)
plt.show()
```



Now increase the duration to 2 seconds ($\Delta f = 0.5\,\text{Hz}$) and recompute.

```
T_long = 2.0
t_long = np.linspace(0, T_long, int(fs * T_long), endpoint=False)
two_sines_long = np.sin(2*np.pi*f1*t_long) + np.sin(2*np.pi*f2*t_long)

X_long = np.fft.fft(two_sines_long)
freq_long = np.fft.fftfreq(len(t_long), 1/fs)[:len(t_long)//2]
mag_long = np.abs(X_long[:len(t_long)//2]) / len(t_long)

plt.figure(figsize=(10, 4))
plt.stem(freq_long, mag_long)# use plot instead of stem for clarity
plt.xlim(5, 15)
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude')
plt.title(f'Two sines at {f1} Hz and {f2} Hz, T = {T_long} s → Δf = 0.5 Hz')
plt.grid(True)
plt.show()
```



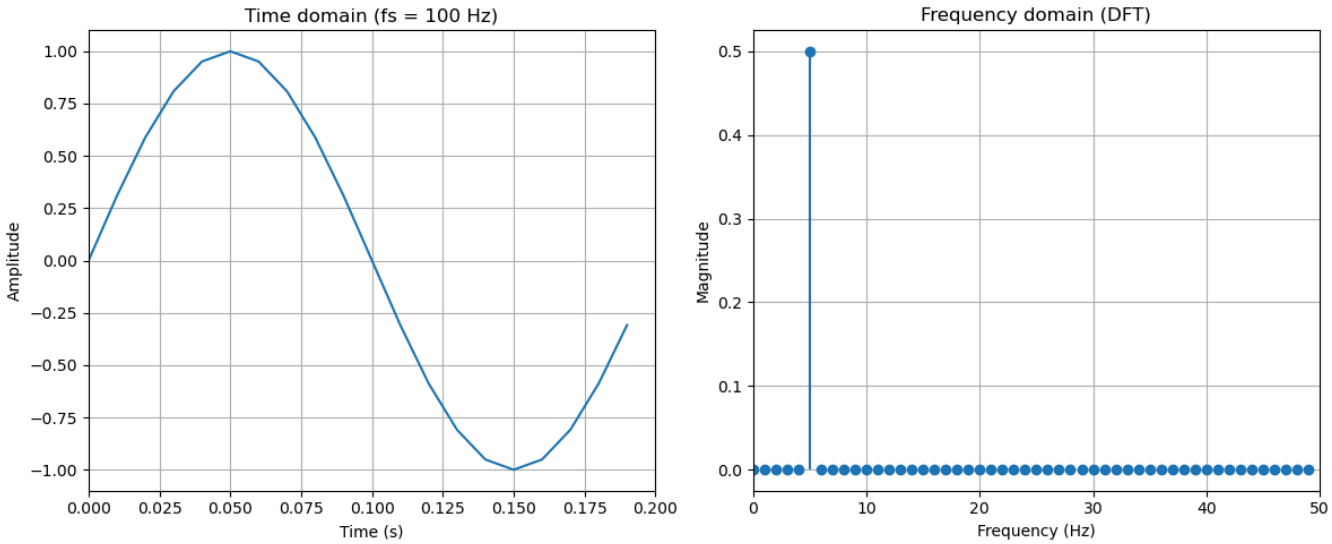
The two peaks are now clearly separated.

Conclusion: To resolve close frequencies, you need a longer observation time.

7. Interactive Demo: Explore Amplitude, Frequency, and Sampling

Use the widget below to vary the frequency of a sine wave and see its DFT update in real time.

```python
def update_plot(frequency=5, amp=1.0, fs=1000):
    # Fixed duration (can be changed if you add another slider)
    T_dur = 1.0
    N_local = int(fs * T_dur)
    t_local = np.linspace(0, T_dur, N_local, endpoint=False)

    # Generate sine wave
    sig = amp * np.sin(2 * np.pi * frequency * t_local)

    # Compute DFT
    X = np.fft.fft(sig)
    freq_axis = np.fft.fftfreq(N_local, 1/fs)[:N_local//2]
    mag = np.abs(X[:N_local//2]) / N_local

    # Plotting
    plt.figure(figsize=(12, 5))

    # Time domain (show first 0.1 seconds, or less if duration is short)
    plt.subplot(1,2,1)
    time_show = min(0.2, T_dur)
    samples_show = int(fs * time_show)
    plt.plot(t_local[:samples_show], sig[:samples_show])
    plt.xlim(0, time_show)
    plt.xlabel('Time (s)')
    plt.ylabel('Amplitude')
    plt.title(f'Time domain (fs = {fs} Hz)')
    plt.grid(True)

    # Frequency domain (up to fs/2)
    plt.subplot(1,2,2)
    plt.stem(freq_axis, mag, basefmt=" ")
    plt.xlim(0, fs/2)
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Magnitude')
    plt.title('Frequency domain (DFT)')
    plt.grid(True)
    plt.tight_layout()
    plt.show()
update_plot(frequency=5, amp=1.0, fs=100)
```




```
# Create interactive widgets
widgets.interact(update_plot,
                 frequency=widgets.FloatSlider(min=1, max=200, step=1, value=5, descriptio
                 amp=widgets.FloatSlider(min=0.1, max=2.0, step=0.1, value=1.0, descriptio
                 fs=widgets.IntSlider(min=50, max=2000, step=10, value=1000, description=
```

Building the Periodic Signal from Its Frequency Components

a periodic signal with period r is composed of r harmonically related frequency components.

Any discrete periodic sequence with period r (where r divides N) can be written as a **Fourier series** with exactly r terms:

$$x[j] = \frac{1}{\sqrt{r}} \sum_{m=0}^{r-1} c_m \, e^{2\pi i \, (m/r) \, j}$$

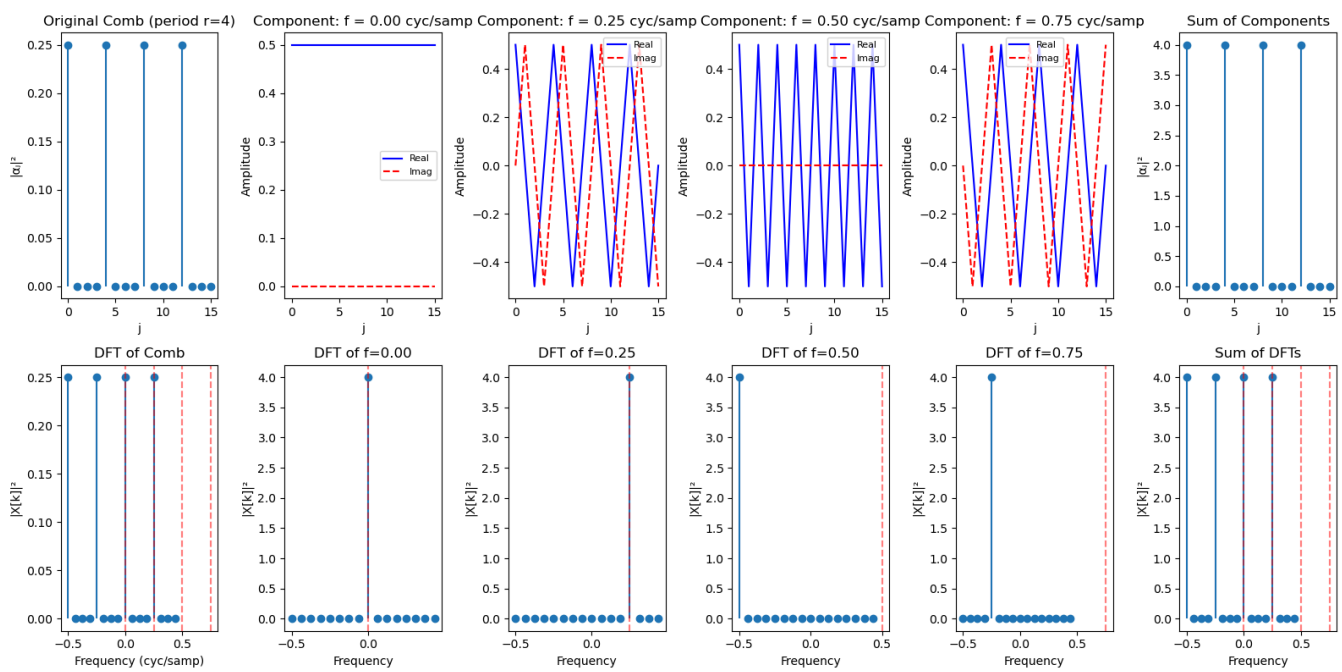
where c_m are the Fourier coefficients. In our case of a uniform comb (all non-zero amplitudes equal), the coefficients are all equal in magnitude, and the DFT shows r peaks of equal height at frequencies $f = 0, \frac{1}{r}, \frac{2}{r}, \ldots, \frac{r-1}{r}$ (cycles per sample).

Thus, the periodic signal **is the sum of r different complex exponentials** (each with its own frequency). The DFT simply reveals those constituent frequencies.

```python
import numpy as np
import matplotlib.pyplot as plt

N = 16
r = 4

# Time indices
j = np.arange(N)

# 1. Create the periodic comb directly (our "quantum state" amplitude pattern)
comb = np.zeros(N, dtype=complex)
for m in range(N // r):
    comb[m * r] = 1.0
comb /= np.linalg.norm(comb)   # normalize

# 2. Build the SAME signal as a sum of r complex exponentials (Fourier series)
freqs = np.arange(r) / r        # frequencies: 0, 1/r, 2/r, 3/r
components = []
reconstructed = np.zeros(N, dtype=complex)
for f in freqs:
    # Complex exponential: e^{2πi * f * j}
    comp = np.exp(2j * np.pi * f * j) / np.sqrt(r)  # normalize each component
    components.append(comp)
    reconstructed += comp

# 3. Compute DFT of the comb and each component
dft_comb = np.fft.fft(comb) / np.sqrt(N)
dft_components = [np.fft.fft(c) / np.sqrt(N) for c in components]

# 4. Plot everything
fig, axes = plt.subplots(2, r+2, figsize=(16, 8))

# Row 1: Time domain (index domain) signals
# Original comb
ax = axes[0, 0]
ax.stem(j, np.abs(comb)**2, basefmt=' ')
ax.set_title(f'Original Comb (period r={r})')
ax.set_xlabel('j')
ax.set_ylabel('|αⱼ|²')
```

```python
# Individual frequency components
for m, (f, comp) in enumerate(zip(freqs, components)):
    ax = axes[0, m+1]
    ax.plot(j, comp.real, 'b-', label='Real')
    ax.plot(j, comp.imag, 'r--', label='Imag')
    ax.set_title(f'Component: f = {f:.2f} cyc/samp')
    ax.set_xlabel('j')
    ax.set_ylabel('Amplitude')
    ax.legend(fontsize=8)

# Reconstructed signal (sum of components)
ax = axes[0, -1]
ax.stem(j, np.abs(reconstructed)**2, basefmt=' ')
ax.set_title('Sum of Components')
ax.set_xlabel('j')
ax.set_ylabel('|αⱼ|²')

# Row 2: Frequency domain (DFT magnitude squared)
freq_axis = np.fft.fftfreq(N)

ax = axes[1, 0]
ax.stem(freq_axis, np.abs(dft_comb)**2, basefmt=' ')
ax.set_title('DFT of Comb')
ax.set_xlabel('Frequency (cyc/samp)')
ax.set_ylabel('|X[k]|²')
for f in freqs:
    ax.axvline(f, color='red', linestyle='--', alpha=0.5)

for m, (f, dft_c) in enumerate(zip(freqs, dft_components)):
    ax = axes[1, m+1]
    ax.stem(freq_axis, np.abs(dft_c)**2, basefmt=' ')
    ax.set_title(f'DFT of f={f:.2f}')
    ax.set_xlabel('Frequency')
    ax.set_ylabel('|X[k]|²')
    ax.axvline(f, color='red', linestyle='--', alpha=0.5)

ax = axes[1, -1]
ax.stem(freq_axis, np.abs(np.sum(dft_components, axis=0))**2, basefmt=' ')
```

```
ax.set_title('Sum of DFTs')
ax.set_xlabel('Frequency')
ax.set_ylabel('|X[k]|²')
for f in freqs:
    ax.axvline(f, color='red', linestyle='--', alpha=0.5)

plt.tight_layout()
plt.show()

print("="*60)
print("EXPLANATION:")
print(f"• The periodic comb with r={r} is exactly equal to the sum of {r} complex exponen
print(f"• These exponentials have frequencies: {freqs} (cycles/sample).")
print(f"• Each exponential produces a single peak at its own frequency in the DFT.")
print(f"• Because the DFT is linear, the DFT of the comb is the sum of the DFTs of these
print(f"• Therefore, the DFT spectrum contains exactly {r} peaks.")
print(f"• Measuring the spacing between adjacent peaks (Δf = 1/r = {1/r:.3f}) reveals the
print("="*60)
```



```
============================================================
EXPLANATION:
• The periodic comb with r=4 is exactly equal to the sum of 4 complex expone
• These exponentials have frequencies: [0.   0.25 0.5  0.75] (cycles/sample)
• Each exponential produces a single peak at its own frequency in the DFT.
• Because the DFT is linear, the DFT of the comb is the sum of the DFTs of t
• Therefore, the DFT spectrum contains exactly 4 peaks.
• Measuring the spacing between adjacent peaks (Δf = 1/r = 0.250) reveals th
============================================================
```